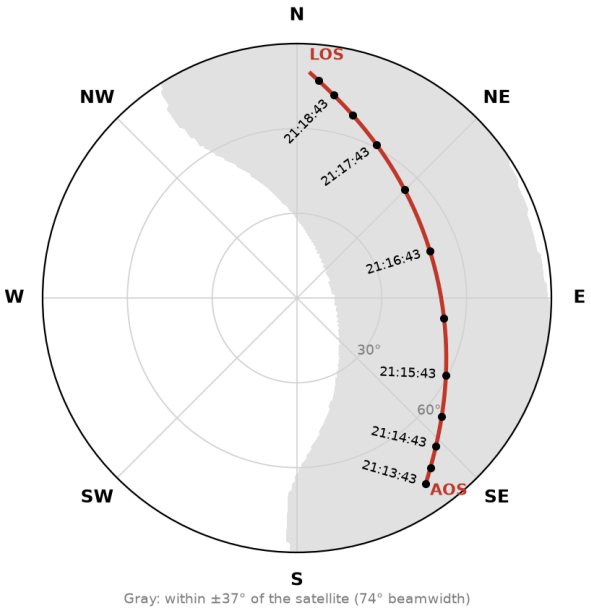
LilacSat-2 Pass Plan

Monday 10/19/2026 Rise 21:13:43 Max 21:16:40 Set 21:19:37 Max El 40° Az 145°→3°

TX (uplink) 144.350 MHz FM, no tone **RX (downlink) 437.200 MHz FM** — Doppler-corrected, rounded to 5 kHz (error up to 2.5 kHz)

Time	TX MHz	RX MHz	Az	El
21:13:43	144.345	437.210	145°	10°
21:14:13	144.345	437.210	142°	14°
21:14:43	144.345	437.210	137°	18°
21:15:13	144.345	437.210	129°	24°
21:15:43	144.350	437.205	117°	31°
21:16:13	144.350	437.205	98°	37°
21:16:43	144.350	437.200	71°	40°
21:17:13	144.350	437.195	45°	36°
21:17:43	144.350	437.195	28°	29°
21:18:13	144.355	437.190	17°	22°
21:18:43	144.355	437.190	10°	17°
21:19:13	144.355	437.190	6°	13°



Gray: within ±37° of the satellite (74° beamwidth)

Yaesu VX-6R + speaker mic: uplink TX

- **Set Mode access:** [F/W] then [0(SET)], dial to the item, [0(SET)] to edit, dial = value, PTT = save
- **Frequency:** VHF band, VFO: key in digits for 144.350; the dial retunes for Doppler
- **Tone:** none: [F/W] then [MODE(SPS) SQ TYP], dial until no TONE shows, PTT to save
- **Power:** [F/W] then [7(LK)TXPO] cycles HIGH / LOW3 (2.5 W) / LOW2 (1 W) / LOW1; use LOW3
- **Set Mode 1 APO:** OFF
- **Set Mode 4 ARS:** OFF
- **Set Mode 6 BCLO:** OFF, so PTT is never blocked by the downlink
- **Set Mode 27 HLF.DEV:** OFF: full deviation
- **Set Mode 37 MCGAIN:** LVL 5, adjust to your voice on the speaker mic
- **Set Mode 51 RPT:** SIMP (simplex); shortcut [F/W] [6(RPT)]
- **Set Mode 67 TOT:** 1 min (default 3)
- **Set Mode 70 TXSAVE:** OFF
- **Do not:** tap [7(LK)TXPO] alone: it turns on the Internet DTMF beep at every TX

Alinco DJ-VX50: downlink RX

- **Menu access:** FUNC, then 8, UP/DOWN to the menu number, dial = value, FUNC = confirm, ESC = exit
- **Frequency:** VFO mode: key in digits, e.g. 437.200 = 4 3 7 2 0 0; the dial then retunes for Doppler
- **Menu 01 R-CDC:** OFF (default): a tone squelch would mute the down-link
- **Menu 03 STEP:** 5K (default is 10K), so each dial click is one ladder step
- **Menu 04 W/N:** WIDE (default)
- **Menu 17 SAVE:** OFF (default is ON): battery saver can clip weak openings
- **Menu 19 SQL:** start at 1-2 (default 9 is tight); raise only if noise is constant
- **Menu 20 REPEAT:** OFF (default), and FUNC 4 until no offset is shown: simplex
- **Keypad lock:** FUNC, then hold #; the dial still retunes, and PTT stays untouched on this radio
- **Audio:** Earphone, volume up enough to hear a weak FM carrier

Notes

- FM transponder planned for 24 h from ~22:00 UTC Mon/Wed/Fri (AMSAT: flexible, subject to change).
- Listen on the downlink first; a telemetry beacon may be heard when the transponder is off.

General

- **Identify:** KF8EEZ, short calls; low power. Keep both radios and the TX antenna well away from the SDR
- **Two radios:** Yaesu VX-6R + MH-57A4B speaker mic = uplink TX (log-periodic); Alinco DJ-VX50 = downlink RX. Use an earphone on the Alinco so its speaker doesn't feed the speaker mic
- **Aim (log-periodic):** point AT satellite: pre-aim at rise Az, follow the Az/El ladder; slant 45°